

Huayi Zheng

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RESEARCH INTERESTS

- North Atlantic heat and freshwater changes
- Earth energy redistribution
- Contemporary sea level change

EDUCATION

Lanzhou University	Sep. 2019 - Jun. 2023
B.S. in Atmospheric Science	
State Key Laboratory of Marine Environmental Science, Xiamen University	Jul. 2022 - Aug. 2022
Visiting student	
Host: Feili Li	
Institute of Marine Sciences of the National Research Council (CNR-ISMAR) in Italy	Nov. 2023 - Dec. 2023
	Nov. 2024 - Dec. 2024
	Aug. 2025 - Sep. 2025
Visiting student	
Host: Andrea Storto	
Institute of Atmospheric Physics, Chinese Academy of Sciences (University of Chinese Academic of Sciences)	Sep. 2023 -present
M.S. in Meteorology	
Advisor: Lijing Cheng	

HONORS AND AWARDS

The First Prize of 14th Student Academic Forum of Harbin Institute of Technology, Harbin Institute of Technology	2021
Outstanding Undergraduate Fellowship, Lanzhou University	2021
National Undergraduate Research Fellowship, China Ministry of Education	2022
Summer Undergraduate Research Fellowship in Marine Environmental Science, State Key Laboratory of Marine Environmental Science(MEL), Xiamen University	2022
National Scholarship, Chinese government	2025

PUBLICATIONS

Pan, Y., Cheng, L., Storto, A., Mayer, M., Gulev, S., Li, Y., Lyu, K., **Zheng, H.** & Yuan, H. (2026). Ocean meridional heat transport estimated from energy budget constraint. Under review in Journal of Climate.

Zheng, H., Cheng, L., Dangendorf, S., Barnoud, A., Trenberth, K., Fasullo, J., & Abraham, J. (2026). Sea level budget in light of recent observational advances since 1960. Under review in Science Advances

Guo, J., Cheng, L., Pan, Y., **Zheng, H.**, & Ma, Z. (2026). Sensitivity of ocean heating rate estimate to time series processing methods. Accepted by Advances in Atmospheric Sciences.

Zheng, H., Cheng, L., Li, F., Pan, Y., & Zhu, C. (2024). An Observation - Based Estimate of Atlantic Meridional Freshwater Transport. Geophysical Research Letters, 51. <https://doi.org/10.1029/2024GL110021>

Cheng, L., Pan, Y., Tan, Z., **Zheng, H.**, Zhu, Y., Wei, W., Du, J., Yuan, H., Guancheng, li, Ye, H., Gouretski, V., Li, Y., Trenberth, K. E., Abraham, J., Jin, Y., Reseghetti, F., Lin, X., Zhang, B., Chen, G., & Zhu, J. (2024). IAPv4 ocean temperature and ocean heat content gridded dataset. Earth System Science Data, 16, 3517–3546. <https://doi.org/10.5194/essd-16-3517-2024>

Zhang, J., **Zheng, H.**, Xu, M., Yin, Q., Zhao, S., Tian, W., & Yang, Z. (2022). Impacts of stratospheric polar vortex changes on wintertime precipitation over the northern hemisphere. Climate Dynamics, 58, 1–17. <https://doi.org/10.1007/s00382-021-06088-x>

SELECTED PRESENTATIONS

General Assembly 2025 of the European Geosciences Union, Vienna, April 2025 (Oral Presentation)
Sea level budget in light of recent observational advances since 1960

The Seventh Xiamen Symposium on Marine Environmental Sciences, Xiamen, January 2025 (Oral Presentation)
Sea level budget in light of recent observational advances since 1960

Joint meeting IQuOD/GTSP/STSP/STOP/XBT science meeting, Bologna, November 2024 (Oral Presentation)
Sea level budget in light of recent observational advances since 1960

The 7th International Ocean Salinity Science Conference, virtual, May 2024 (Oral Presentation)
An observation-based estimate of the Atlantic meridional freshwater transport from 2004 to 2012

General Assembly 2024 of the European Geosciences Union, Vienna, April 2024 (Poster)
An observation-based estimate of the Atlantic meridional freshwater transport from 2004 to 2012

Marine Multidisciplinary Cross-Integration Seminar, Xiamen, July 2022 (Oral Presentation)

Quantifying the multiple time-scales variability of salinity amplification

Asia-Oceania Geoscience Society (AOGS) 18th Annual Meeting, virtual, August 2021 (Oral Presentation)

Global and regional salinity changes related to ENSO

COMMUNITY ACTIVITIES

Member, Asia Oceania Geosciences Society

Aug 2021 – Aug 2022

Member, European Geosciences Union

April 2024 – present

SKILLS

- **Languages:** Fluent - English and Native - Chinese
- **Computer:** Python, NCL, Fortran. LATEX